Kayak LED Turret — Bench Bring-Up Checklist

Firmware: `kayak_working/kayak_working.ino` · Netlist: `kayak working.txt` (KiCad 9.0.6 netlist, rev 2.0)

DATE	TESTER	PCB REV / SERIAL	FIRMWARE COMMIT
		rev 2.0 /	d4c19c1
NANO ESP32 ID	TMC MODULE A2 (LIFT)	TMC MODULE A3 (PAN)	12 V SUPPLY
	R10:	R10:	

Tick each box when the check passes. Write DMM readings and anything unexpected in Reading / notes. Stop at the first failure.

Work top to bottom. Do not start a stage until the previous one passes.

Standalone bench sketches (`bench_tests/`, see its README) test one subsystem each and are the quickest way through the early stages:

Stage	Flash this first	Then
3 (first boot on the PCB)	`test_buttons` (USB power is enough), then `test_imu` (needs 12 V, and the Stage 2.4 pull-up check passed), then `test_tmc_uart` (never enables a driver)	`kayak_working` boot checks
4	`test_motor`, `TEST_AXIS` 0 then 1	`kayak_working` jog / chord / limit checks
5	`test_led_strip`, `TEST_STRIP` 0–3	`kayak_working` budget checks
6	`test_lightbar`	`kayak_working` lift-gated light bars
7–8	—	`kayak_working` IMU / tracking

Never plug or unplug a stepper motor while 12 V is on (TMC2209 output stage damage).

Open items: resolve before or during bring-up

Closed	#	Item	Blocks	How to resolve
☐	1	**TMC UART wiring: D1/TX reaches nothing on factory modules**	Stage 3 green LED, all motion	See Open item 1
☐	2	**GY-521 I²C pull-up rail** (3.3 V or 5 V?)	Stage 2 (before the Nano is fitted)	Measure in Stage 2, step 2.4
☐	3	**Level shifter part**: netlist value says `74HC245`, build uses SN74HCT245N	Stage 5 strips	Read the chip marking. It must say **HCT**. An HC part at 5 V needs VIH ≥ 3.5 V, and 3.3 V drive is out of spec
☐	4	**Unidentified pot-like part on the TMC2209 module**	Nothing (UART mode ignores it). See note	See Open item 4
☐	5	**No homing yet**: pan must be at full LEFT and lift at full DOWN at power-on	Stage 4 soft limits	Position by hand before every power-on until StallGuard homing is tuned
☐	6	**Outputs float before `setup()`** (ROM bootloader + ~0.9 s core double-tap fade): LED data inputs on U1 and D8 are undriven	Board revision	See Board revision items

Stage 0: Build and flash (Nano off the board, USB only)

- [ ] **1.** Arduino IDE → Board **Arduino Nano ESP32**, core **esp32 2.0.17**.
- [ ] **2.** **Tools → Pin Numbering → "By GPIO number (legacy)"**. The sketch has an `#error` guard and will not compile otherwise. CLI: `arduino-cli compile kayak_working` (reads `kayak_working/sketch.yaml`).
- [ ] **3.** Libraries: TMCStepper 0.7.3, AccelStepper ≥ 1.64, Adafruit NeoPixel 1.15.5, Adafruit MPU6050 2.2.9, Adafruit Unified Sensor 1.1.15, Adafruit BusIO 1.17.4.
- [ ] **4.** **Do not flash the old firmware onto this board.** Its EN/DIR assignments are swapped, and it runs I²C on A6/A7, which are the DIAG nets.
- [ ] **5.** Flash the bare Nano over USB and open the Serial Monitor at 115200.

✓	Check	Pass	Reading / notes
☐	RGB LED at boot	**Blue** briefly, then **red** (no drivers present)	
☐	Serial	`[TMC] UP/DOWN (A2, addr 0): test_connection = 2 (NO REPLY ...)`, same for LEFT/RIGHT, `driver config FAILED ... left DISABLED`, `MPU6050 NOT DETECTED`, `Setup complete`	
☐	`?`	Menu lists `? e d r s i b p t`	
☐	`e`	`refused: config not verified` for both	
☐	DMM: D2 and D5 to GND	**3.3 V** (~EN HIGH = disabled)	
☐	DMM: D8 to GND	**0 V** (light-bar PWM 0)	

Stage 1: Unpowered continuity (no Nano, no modules, no supply)

Every row is a netlist connection. Beep it out on the bare PCB.

✓	Nano pin	GPIO	Net	Destination
☐	D0/RX	44	`Net-(A1-D0/RX)`	A2 pin 5 **and** A3 pin 5 (module "RX" = UART_RX = PDN_UART)
☐	D1/TX	43	`Net-(A1-D1/TX)`	A2 pin 4 **and** A3 pin 4 (module "TX" = UART_TX)
☐	D2	5	`Net-(A3-~{EN})`	A3 pin 8 EN (pan)
☐	D3	6	`Net-(A3-STEP)`	A3 pin 2 STEP (pan)
☐	D4	7	`Net-(A3-DIR)`	A3 pin 1 DIR (pan)
☐	D5	8	`Net-(A2-~{EN})`	A2 pin 8 EN (lift)
☐	D6	9	`Net-(A2-STEP)`	A2 pin 2 STEP (lift)
☐	D7	10	`Net-(A2-DIR)`	A2 pin 1 DIR (lift)
☐	D8	17	`Net-(J2-Pin_5)`	J2.5 PWM CONTROL (direct, not level-shifted)
☐	D9	18	`Net-(U1-A3)`	U1 pin 5 (A3) → B3 pin 15 → R4 → J2.4 Cockpit Back
☐	D10	21	`Net-(U1-A2)`	U1 pin 4 (A2) → B2 pin 16 → R3 → J2.3 Cockpit Front
☐	D11	38	`Net-(A1-D11_MOSI)`	U1 pin 3 (A1) → B1 pin 17 → R2 → J2.2 Bottom Right
☐	D12	47	`Net-(A1-D12_MISO)`	U1 pin 2 (A0) → B0 pin 18 → R1 → J2.1 Bottom Left

✓	Nano pin	GPIO	Net	Destination
☐	A0	1	`Net-(J6-Pin_4)`	J6.4 RIGHT
☐	A1	2	`Net-(J6-Pin_3)`	J6.3 LEFT
☐	A2	3	`Net-(J6-Pin_2)`	J6.2 MODE
☐	A3	4	`Net-(J6-Pin_1)`	J6.1 BRIGHTNESS
☐	A4/SDA	11	`Net-(A1-A4/SDA)`	GY-521 SDA
☐	A5/SCL	12	`Net-(A1-A5/SCL)`	GY-521 SCL
☐	A6	13	`Net-(A1-PadA6)`	A2 DIAG (lift)
☐	A7	14	`Net-(A1-PadA7)`	A3 DIAG (pan)
☐	VIN	—	`+12V`	J5.1, A2/A3 VS (pin 9)
☐	3V3	—	`+3V3`	A2/A3 VIO (pin 15), A3 MS1 (pin 7)
☐	GND	—	`GND`	J1.2, J5.2, J6.5, U1 pins 10 and 19 (CE), A2/A3 GND pins, A2 MS1+MS2, A3 MS2, GY-521 GND
☐	—	—	`+5V`	J1.1, U1 pin 20 (VCC), U1 pin 1 (DIR "A→B"), GY-521 VCC

✓	Check	Pass	Reading / notes
☐	Every row above	Beeps	
☐	+12V↔GND, +5V↔GND, +3V3↔GND	**No** short (expect a cap-charging blip, then open)	
☐	D13	Connected to nothing	
☐	U1 chip marking	**SN74HCT245N** (open item 3)	

Stage 2: Power rails (12 V on, no Nano, no modules, no motors, no strips)

✓	Step	Probe (DMM, black on GND)	Pass	Reading / notes
☐	2.1	J5.1 / Nano VIN socket pin	**11.5–12.6 V**	
☐	2.2	J1.1 / U1 pin 20 / GY-521 VCC	**4.9–5.2 V**	
☐	2.3	U1 pin 1 (DIR)	= 5 V rail (HIGH, A→B) · U1 pin 19 (CE) = **0 V**	
☐	2.4	**GY-521 fitted, Nano still out**: SDA and SCL to GND	**≈ 3.3 V → OK. ≈ 5 V → STOP**: pull-ups go to VCC, which would put 5 V on A4/A5. Remove the GY-521's pull-ups or add a level shifter before fitting the Nano	
☐	2.5	Module sockets VS (pin 9)	**12 V** · VIO (pin 15) = **0 V** (no Nano yet)	

Stage 3: Nano fitted, modules fitted, 12 V on, motors and strips disconnected

✓	Check	Pass	Reading / notes
☐	Module VIO (pin 15), both	**3.3 V**	

✓	Check	Pass	Reading / notes
☐	Module VS (pin 9), both	**12 V**	
☐	RGB LED	**Blue** (≈ 0.1 s) → **green** if open item 1 is fixed, **red** if not	
☐	EN pins (module pin 8)	**0 V** if that driver verified (green), **3.3 V** if it failed. A failed driver must stay at 3.3 V	
☐	UART idle, D1	**3.3 V** (TX idles HIGH)	
☐	UART idle, D0	**≈ 3.0 V** with 1 kΩ fitted (the modules' 2 × 20 kΩ pull-downs against the 1 kΩ). **≈ 0 V means TX is not reaching PDN_UART** (R10 still open)	
☐	DIAG, A6 and A7	**0 V**	
☐	D8	**0 V** (light bars off: lift is not at its UP limit)	

Boot log when it passes:

```
[TMC] UP/DOWN (A2, addr 0): writes 21/21 acked, GCONF ok, CHOPCONF ok, PWMCONF ok → CONFIG OK (IRUN set 795 mA RMS,
IHOLD 30%)
[TMC] LEFT/RIGHT (A3, addr 1): writes 21/21 acked, ... → CONFIG OK ...
[TMC] both drivers verified and ENABLED
MPU6050 OK -- calibrating gyro bias, keep still ~1 s
```

`d` must show, for **each** driver: `test_connection : 0 (OK)`, `config : VERIFIED (IFCNT 21/21 at boot)`, `~EN pin : LOW (enabled)`, `microsteps=16`, `rms_current(set)=795 mA`, `stealth=1`, `stst=1`, `ot=0 otpw=0 s2ga=0 s2gb=0`. `ola=1 olb=1` (open load) is **expected with no motor connected**.

`d` takes about 0.1–0.3 s (many UART reads). The buttons are not sampled during that time, so don't run it while jogging.

Fail modes: `test_connection = 2 (NO REPLY)` → open item 1 (TX path), wrong address, or no 12 V on VS (the TMC2209 logic is powered from VS). `= 1 (0xFFFFFFFF)` → bus stuck HIGH. `writes n/21` with n < 21 → UART bit errors (check the series resistor value and wiring).

Stage 4: Motors (12 V OFF while connecting!)

Hand-position pan at **full LEFT** and lift at **full DOWN** (open item 5), connect J3 (lift) and J4 (pan), then power on.

✓	Check	Pass	Reading / notes
☐	`d`	`ola=0 olb=0` with the motor connected; `cs_actual` ≈ **7** at standstill (IHOLD ≈ 30 % of IRUN 25)	
☐	`s`	Each axis moves out 200 steps and back, then `[s] test jog complete`. During the move `cs_actual` ≈ **25**; ≈ 0.5 s after stopping it falls back to ≈ **7** (automatic standstill reduction)	
☐	Hold LEFT / RIGHT	Pan moves left / right, decelerates smoothly on release	
☐	Hold at a limit	Motion stops at position 0 / `LR_STEP_LIMIT`, never past it (`d` → `position`)	
☐	BRIGHTNESS + RIGHT / + LEFT	Lift up / down; **no pan**, no brightness change	
☐	Release BRIGHTNESS first, keep RIGHT held	Nothing moves until RIGHT is released and pressed again	
☐	Mark the pan at 0 and at `LR_STEP_LIMIT`	Arc **≤ 180°**. If larger, lower `LR_STEP_LIMIT`	
☐	Wrong direction	Swap one coil pair at J3/J4, or set the DIR polarity in firmware	

✓	Check	Pass	Reading / notes
☐	Motor case after 5 min holding	Warm, not too hot to touch (< ~60 °C)	
☐	`e`	Both disable (EN 3.3 V, motors go limp; **the lift may drop under load**, and the firmware's position is then wrong: re-position by hand), `e` again re-enables	
☐	**12 V dropout with USB still connected**: switch the 12 V supply off for 2 s, then back on	Within 0.25 s: `[TMC] ... FAULT: no reply or config lost ... DISABLED`, LED **red**, EN pins **3.3 V**, and they **stay** 3.3 V when 12 V returns (the motors must not energise). Then `r` → green, EN 0 V	

Stage 5: LED strips (strip 12 V supply on, data via J2.1–J2.4)

✓	Check	Pass	Reading / notes
☐	Boot	All strips dark until `Setup complete`, then nav lights at the lowest level	
☐	`b` at boot	`lit 600/600 (1.000) x applied 16/255 = 0.0627`	
☐	Click BRIGHTNESS to the top level, `b`	`requested 220 ... applied 51 ... = 0.2000 ** LIMITED **`	
☐	Brightness levels in all-lit modes (nav, flood, pulse)	**Expected:** levels 2, 3 and 4 (requested 84/152/220) all show at applied 51, so only level 1 (16) looks different. That is the 0.20 budget doing its job, not a button fault. If you want four visible steps there, the levels have to sit at or below 51	
☐	MODE through every mode, `b` each time	`product` **≤ 0.2000** always	
☐	Strip supply current (clamp meter), white flood at top level	Record it. It must not exceed ≈ 20 % of the strips' rated full-white current	
☐	Colour order	Bottom Left red, Bottom Right green, cockpit white. If wrong, change `NEO_GRBW` ↔ `NEO_RGBW`	

Stage 6: Light bars (MOSFET module on J2.5)

✓	Check	Pass	Reading / notes
☐	`p 64`	D8: DMM DC ≈ **0.83 V**, frequency (DMM Hz mode) ≈ **5.0 kHz**; bars dim	
☐	`p 255` / `p 0`	≈ 3.3 V / 0 V	
☐	`p auto`, lift not at UP	Bars **off**	
☐	Lift to its UP soft limit (BRIGHTNESS + RIGHT until it stops)	Bars **on** at the brightness level's duty (64/128/191/255); they go off as soon as the lift moves down	

Stage 7: IMU

✓	Check	Pass	Reading / notes
☐	`i` within 1 s of boot	`calibrating gyro bias (keep still)`	
☐	`i` flat and still	roll and pitch ≈ 0 ± 2°, `errors 0`, `samples` increasing ≈ 100/s	

✓	Check	Pass	Reading / notes
☐	Rotate the board 90° by hand, slowly	heading changes by ≈ 90° (sign = mounting)	
☐	Still for 5 min	heading drift: record it (expect a few °/min; this is why re-zeroing exists)	
☐	Hold MODE ≥ 1 s	`[MODE long] heading re-zeroed`, lighting mode **unchanged**	
☐	Tilt 30°	roll/pitch follow within ~1 s, no runaway	

Stage 8: Pan tracking (scaffold, off by default)

- ☐ **1.** `t` → `pan tracking ENABLED`.
- ☐ **2.** Rotate the base slowly: the pan should counter-rotate to hold its world direction. **If it turns the same way, set `PAN_TRACK_SIGN = -1.0f`.**
- ☐ **3.** Hold LEFT: manual pan wins. Release: tracking resumes from the new position.
- ☐ **4.** At a soft limit, tracking stops there (it never drives past it).
- ☐ **5.** `t` again to disable. Leave `g_panTrackEnabled = false` in the code until this stage passes on the water.

What "done" looks like

- ☐ Green LED at every boot, `d` clean on both drivers.
- ☐ All Stage 4–7 checks pass, with `b` product ≤ 0.20 in every mode and level.
- ☐ `LR_STEP_LIMIT` arc measured at ≤ 180°.
- ☐ Open items 1–3 closed, 4 photographed and measured, 5 either accepted (manual positioning) or replaced by StallGuard homing (tune `TMC_TCOOLTHRS` / `TMC_SGTHRS`).

BRING-UP COMPLETE: NAME	SIGNATURE	DATE

Open item 1: TMC2209 UART TX path (R10)

The BTT TMC2209 V1.3 module schematic (`TMC2209 V1.3SCH.pdf`) and manual show:

- Header pin 5 **"RX" (UART_RX)** connects **directly** to the chip's PDN_UART pin, with a 20 kΩ pull-down (R6).
- Header pin 4 **"TX" (UART_TX)** connects to PDN_UART **only through R10, which is not fitted** at the factory ("UART (default)" = RX pin only). The manual's "DIY" option is to **solder R10's pads together**.
- **There is no series resistor on the module.**

This PCB wires Nano D0/RX → both modules' RX pins and Nano D1/TX → both modules' TX pins. With factory modules, **D1 drives nothing**: the drivers never receive a command, `test_connection` returns `NO REPLY`, the LED goes red and both drivers stay disabled. That is a safe failure, but nothing moves.

Option	Verdict
A. Fit a 1 kΩ resistor at R10 on ONE module (match the pad size)	**Recommended.** Gives the standard single-wire topology: D1 —1 kΩ— PDN_UART bus — D0. Works with the firmware as-is. Fitting 1 kΩ on **both** modules is also acceptable (500 Ω effective; ≈ 6.6 mA when a driver pulls the bus low), and keeps the modules interchangeable
B. Solder-bridge R10 (as the BTT manual shows)	**Do not.** That puts D1's push-pull output directly on PDN_UART. Every time a driver answers, it drives LOW against D1's idle HIGH with no current limit. Firmware cannot prevent this
C. Next PCB revision	Put a 1 kΩ in series from D1 to the UART_RX net on the carrier board and leave the module TX pins unconnected. Firmware unchanged

Also note: the ESP32-S3 ROM prints its boot log on GPIO43 (D1) at every reset. With option A it reaches the PDN_UART bus. The TMC2209 rejects it (no valid sync/CRC), and both drivers are disabled through boot anyway (EN has a 20 kΩ pull-up to VIO).

Open item 4: the pot-like part on the TMC2209 V1.3

It is almost certainly the **VREF trimmer**: the small square trimpot with a cross slot next to the DIAG and VS pins, visible in the manual's top-view photo. The module schematic draws VREF (chip pin 17) as a fixed 20 kΩ/20 kΩ divider from 5VOUT (R8/R9) plus R11 = 0 Ω, so the trimpot replaces part of that divider on the real board.

What to capture so it can be confirmed:

1. Photos: the top side square-on, a close-up of the part's marking (e.g. `103` = 10 kΩ, `203` = 20 kΩ), and the bottom side around R8, R9 and R11.
2. Module **unplugged**: resistance between the two outer terminals (fixed value), then wiper to each end while turning it (should sweep 0 → full).
3. Module plugged in, 12 V on, motors disconnected: DC voltage from the wiper to GND while turning it slowly. Expect about **0 → ~2.5 V** (VREF range).
4. Continuity from the wiper to R11 (the 0 Ω link to the VREF pin).

Why it does not matter in this firmware: `I_scale_analog` is written **0** and read back, so the current comes from the internal reference and IRUN/IHOLD only. The trimpot setting has no effect in UART mode. It would only set the current if a module ran standalone, and the firmware never enables a driver whose UART config failed. Still turn it to mid-range so that a module moved into another, standalone machine starts at a sane current.

Board revision items

- **Pull-downs on the LED data inputs and the PWM output.** 10 kΩ to GND on U1 A0–A3 (pins 2–5) and on J2.5 / D8. Until `setup()` runs (ROM bootloader, then the core's ~0.9 s double-tap RGB fade), these pins float. Floating HCT inputs can clock random data into the SK6812s, at any brightness and outside the power budget, and can half-turn-on the light-bar MOSFET. Firmware cannot run earlier than this. Until the board is revised, power the strips **after** the Nano has booted (RGB LED green or red).
- **Tie the unused 74HCT245 inputs A4–A7 (pins 6–9) to GND.** They are floating CMOS inputs.
- **UART:** add a 1 kΩ in series from D1 to the UART_RX net, and leave the module TX pins unconnected (see Open item 1, option C).